

Significance, and the unit you are counting

- BIB — Bioinformatics for Bioengineering / Biyomühendislik için Biyoinformatik
- Week 6 · Block B "Is it real?"
- Anlamlılık ve neyi saydığınız: istatistiksel bağımsızlık birimi
- This is the load-bearing lecture of the course. Four later weeks cash it.

Ad of the Week #6

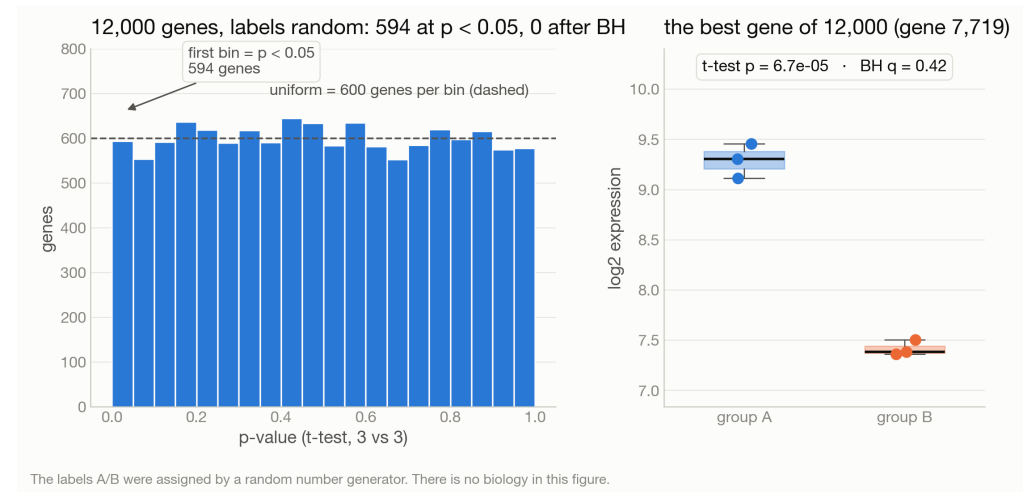
- Skills Ledger tick: "statistical inference"

Two groups. Nothing is happening. Commit to a number.

- 12,000 genes. Six samples. All six drawn from the same distribution.
- Then I label three of them "A" and three of them "B", at random
- There is no biology here. There is no treatment. There is no difference. I made the labels up.
- I run a t-test on every gene and count how many have $p < 0.05$
- Before I press run: write a number on your paper. Out loud, someone, commit.

The answer is about 600. Every time.

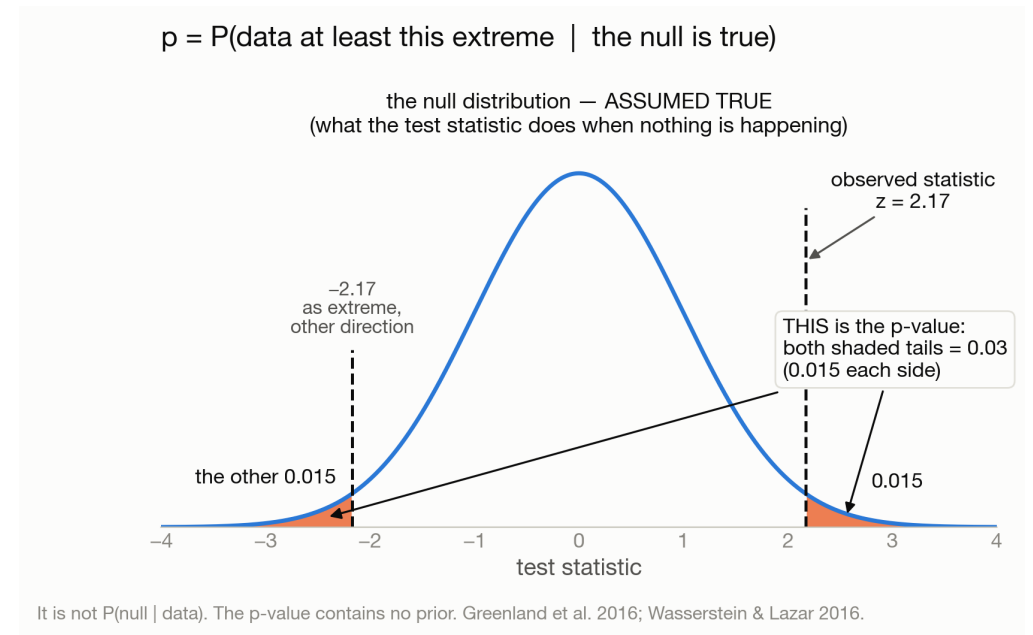
- $12,000 \text{ genes} \times 0.05 = 600$ — that is what $p < 0.05$ means
- Your top gene will have a p-value of maybe $1e-5$ and a beautiful boxplot
- After Benjamini–Hochberg correction: typically 0
- The p-value histogram is flat — uniform on $[0,1]$, a true null
- A big spike near zero is real signal. Flat means nothing is there.
- This whole lecture is contained in this figure.



12,000 genes, six samples from one distribution, labels random: 594 genes at $p < 0.05$, 0 after BH — and the best gene still has a beautiful boxplot ($p = 6.7e-05$).

What a p-value is

- Operational definition, and this is the only one:
 - The probability of observing data at least this extreme, IF the null hypothesis were true
- It is a statement about data, conditional on a hypothesis
- It is not a statement about the hypothesis, conditional on data
- $p = 0.03$ means: "if nothing were going on, I would see something this extreme 3% of the time"
- That is genuinely all it says. Everything else people claim it says is a mistake.



$p = P(\text{data at least this extreme} \mid \text{null true})$: both shaded tails together are 0.03 of the curve that is assumed true — 0.015 on each side of ± 2.17 .

Four sentences you must never write

-  "p = 0.03, so there is a 97% chance the null is false"
— the p-value contains no prior
-  "p = 0.001, so the finding is definitely real" — small p ≠ true; it means surprising under the null
-  "p = 0.001, so the effect is large" — p is not an effect size (etki büyüklüğü). It confounds effect and n.
-  "p = 0.04, so this will replicate" — p is a poor predictor of replication
-  What you write instead: the effect size, its confidence interval, the n, and the correction applied
- I will delete these four sentences from your notebooks all semester. Save us both the time.

Four sentences you must never write

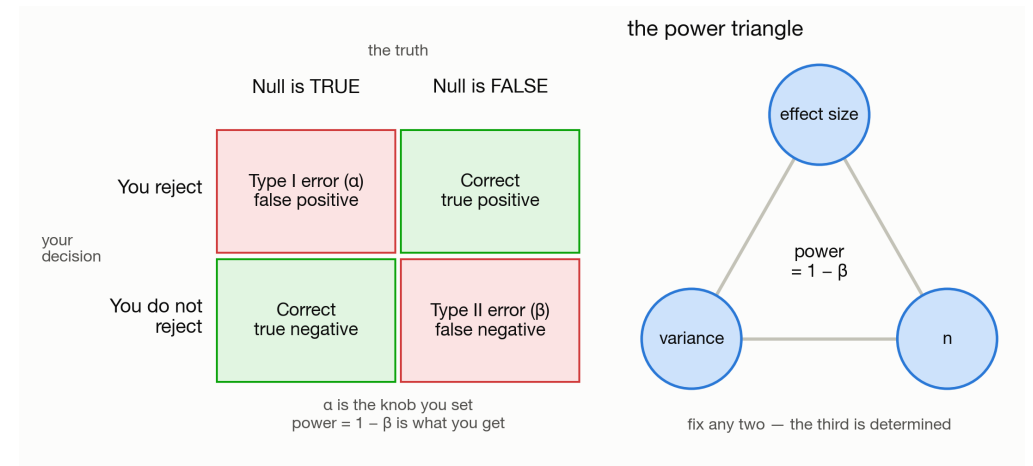
-  ~~"p = 0.03, so there is a 97% chance the null is false"~~
the p-value contains no prior
-  ~~"p = 0.001, so the finding is definitely real"~~
small p = surprising under the null, not true
-  ~~"p = 0.001, so the effect is large"~~
p confounds effect size and n
-  ~~"p = 0.04, so this will replicate"~~
p is a poor predictor of replication

 What you write instead:
the effect size, its confidence interval, the n, and the correction applied

Four sentences you must never write — and what you write instead.

Type I, Type II, and power

- Type I (α) = false positive · Type II (β) = false negative
- Power (güç) = $1 - \beta$ = the probability of detecting a real effect that is genuinely there
- The power triangle: effect size $\leftrightarrow$ variance $\leftrightarrow$ n. Fix any two, the third is determined.
- α is the knob you set. Power is what you get. Most biologists set α and never compute power.

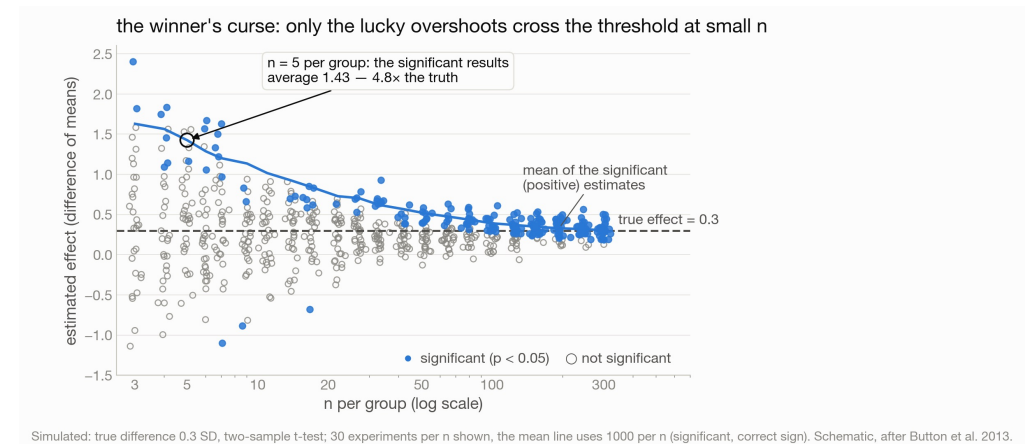


Type I (α) and Type II (β) errors, and the power triangle: effect size, variance, n.

	Null TRUE	Null FALSE
Reject	Type I (α)	Correct
Do not reject	Correct	Type II (β)

The winner's curse: underpowered studies do not just miss things

- The intuition: an underpowered study might miss a real effect. Not the danger.
- The dangerous part: significant results are systematically exaggerated
- Why: only the samples that overshoot cross the threshold — you filtered on noise
- So the published effect is biased upward, and the follow-up "fails to replicate"
- It did not fail to replicate. The first study over-estimated it.
- Small n is not conservative — it is unreliable in both directions



The winner's curse: at $n = 5$ per group the significant results average 1.43 against a true effect of 0.3 — 4.8x the truth. Schematic, after Button et al. 2013.

THE UNIT OF INDEPENDENCE — the load-bearing block

- Every test you run needs a number, n . Getting that number wrong invalidates everything downstream.
- The rule. Write this down verbatim:
 - n is the number of units you could have replaced with a different draw from the population you want to talk about.
- Not the number of rows in your table. Not the number of measurements. Not the number of cells.
- Three cases coming. The room votes on n for each. I want hands up, not silence.

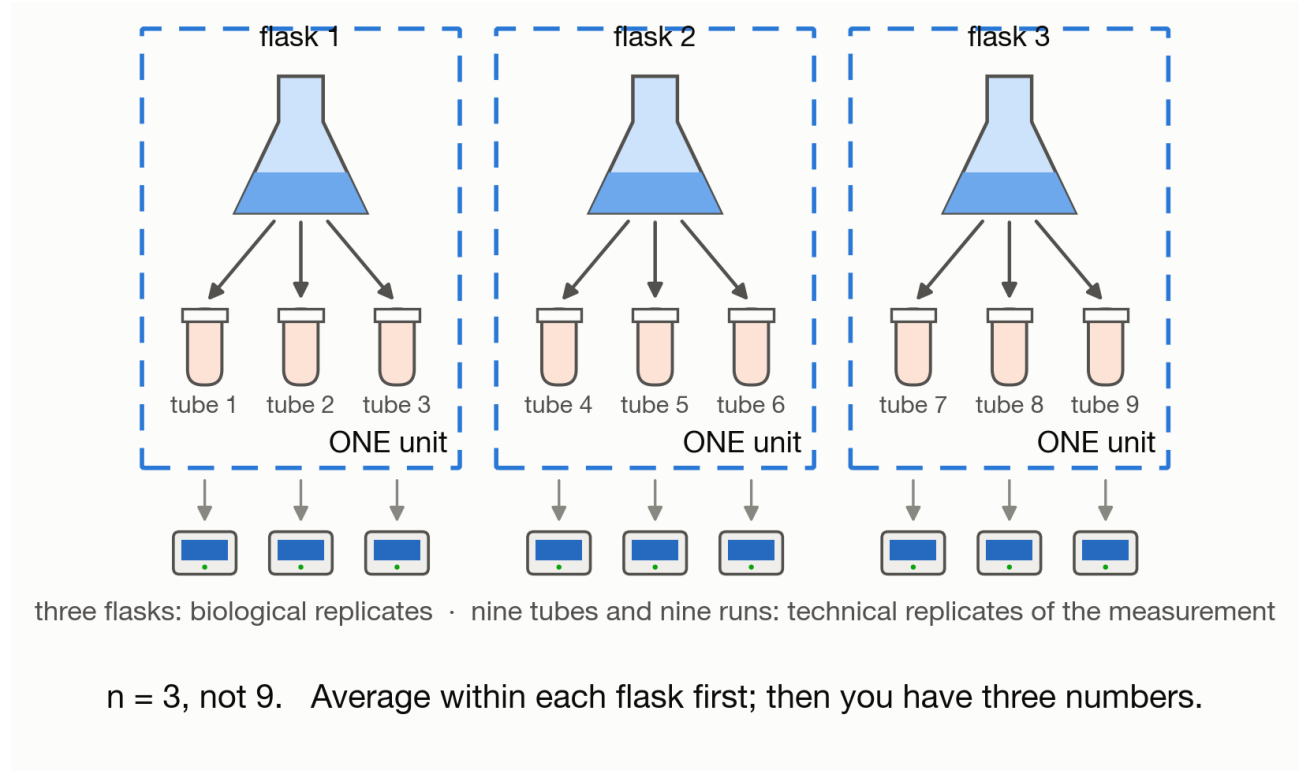
The unit of independence

n is the number of units you could have replaced with a different draw from the population you want to talk about.

Not rows. Not measurements. Not cells.

Case 1: three flasks, nine tubes

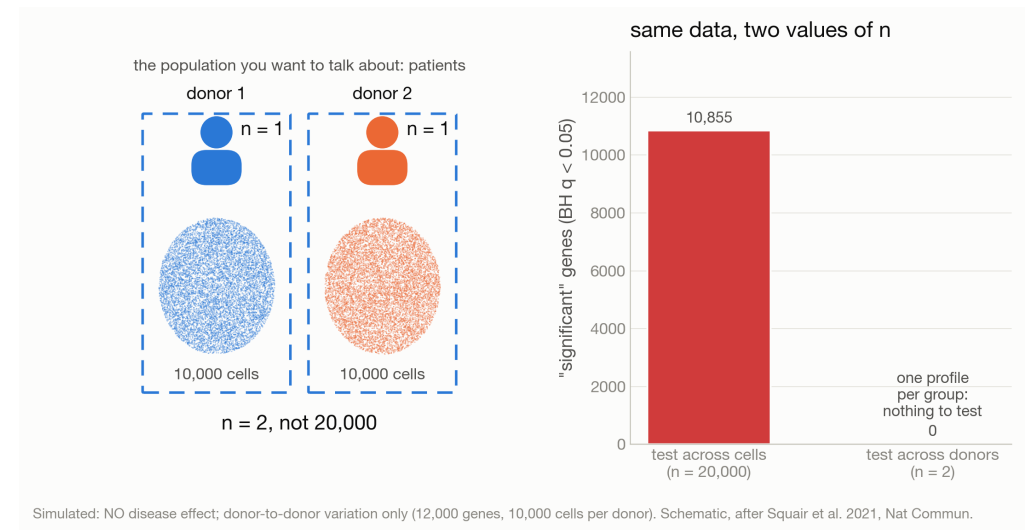
- You grow bacteria in three independent flasks under one condition
- From each flask you take three aliquots and prepare three libraries
- You sequence all nine
- Your conclusion will be about this condition
- Vote: is $n = 9$, or $n = 3$?



Three flasks, nine tubes, nine runs: each flask with its three tubes is ONE unit. $n = 3$, not 9.

Case 2: 10,000 cells, two donors

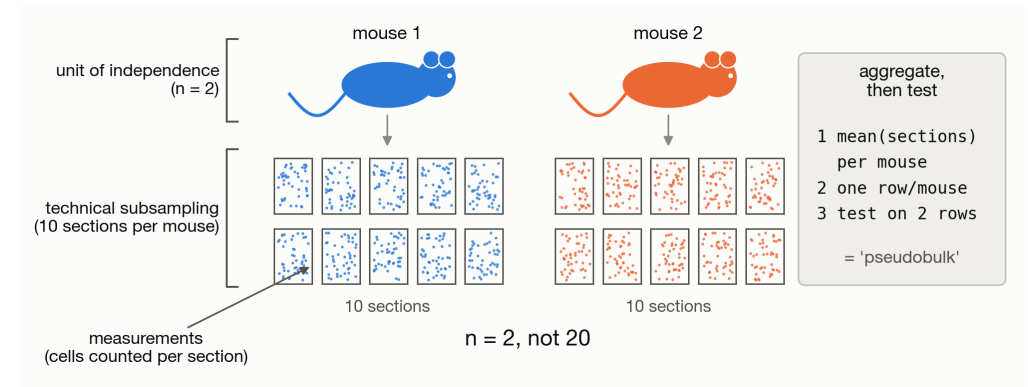
- Single-cell RNA-seq. Two donors. You recover 10,000 cells from each.
- You compare gene expression between the two donors' cells
- Your conclusion will be about this disease
- Vote: $n = 20,000$, or $n = 10,000$ per group, or $n = 2$?
- This is the most common error in the published single-cell literature. It returns in Week 10.



Two donors, 10,000 cells each: $n = 2$. Simulated with NO disease effect, testing across cells calls 10,855 genes 'significant'; across donors there is nothing to test. Schematic, after Squair et al. 2021.

Case 3: twenty sections, two mice

- Immunohistochemistry. Two mice, ten sections each. You count cells per section.
- Vote: $n = 20$, or $n = 2$?
- Same answer as case 2, different technique — and this one you have probably done
- The general form: measurements NESTED inside a unit — the unit is what you count
- Fix (a): aggregate to the unit, then test on the aggregate
- Fix (b): a mixed-effects model with the unit as a random effect
- (a) is almost always right for this course — it is what pseudobulk is



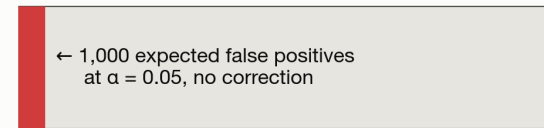
Measurements nested in sections nested in mice: the unit is the mouse, $n = 2$, not 20. Aggregate to the unit, then test — that is what pseudobulk means.

Multiple testing: 20,000 tests, before biology starts

- One test at $\alpha = 0.05$: a 5% chance of a false positive. Acceptable.
- 20,000 tests at $\alpha = 0.05$: about 1,000 false positives, guaranteed, before biology enters the room
- $P(\text{at least one false positive among 20 independent tests}) = 1 - 0.95^{20} = 0.64$

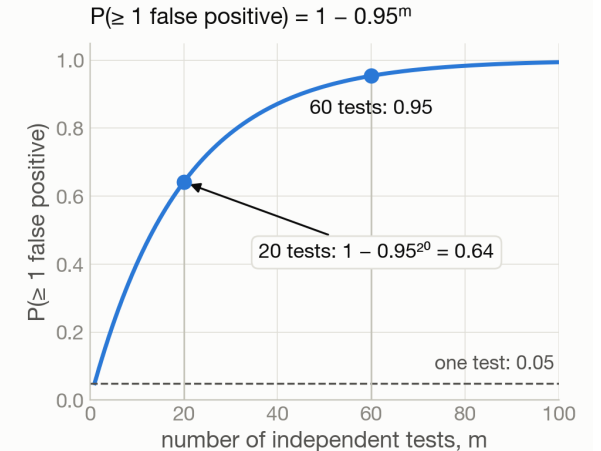
20,000 tests at $\alpha = 0.05$: about 1,000 false positives before biology enters

one bar = 20,000 genes tested



an expectation, not a promise: $20,000 \times 0.05 = 1,000$ on average

0 5,000 10,000 15,000 20,000
genes tested



Bonferroni: test each at α/m (controls FWER). Benjamini-Hochberg 1995: controls the expected fraction of false rejections (FDR).

20,000 tests at $\alpha = 0.05$: about 1,000 expected false positives before biology enters; $P(\geq 1 \text{ false positive}) = 1 - 0.95^m$, 0.64 at $m = 20$, 0.95 at $m = 60$.

FWER or FDR: which promise do you want?

- FWER — family-wise error rate: the probability of any false positive
- Bonferroni: test at α/m . One line. Brutally conservative.
- FDR — false discovery rate: the expected proportion of your rejections that are false
- Benjamini–Hochberg, 1995 — the genomics default
- For genomics FDR is almost always the right currency: you accept that a small fraction of your hits are wrong

BH, by hand, once

- Ten p-values, sorted. $m = 10$, $\alpha = 0.05$. BH critical value for rank i is $(i/m) \cdot \alpha$
- Find the largest i that passes — here $i = 6$ — and reject ranks 1 through 6
- Bonferroni at $0.05/10 = 0.005$ would have rejected only 2

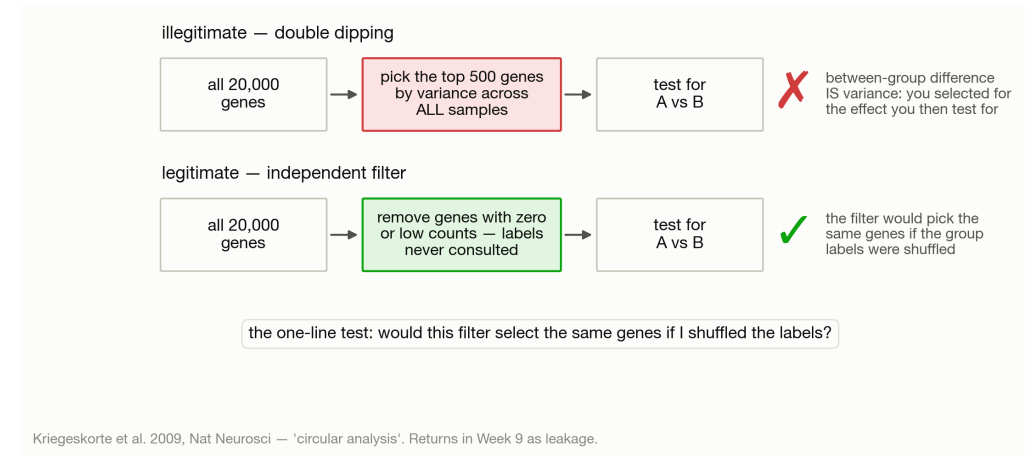
i	p	$(i/10) \cdot 0.05$	$p \leq \text{crit?}$	q
1	0.0001	0.005	✓	0.0010
2	0.0008	0.010	✓	0.0040
3	0.0071	0.015	✓	0.0225
4	0.0090	0.020	✓	0.0225
5	0.0130	0.025	✓	0.0260
6	0.0250	0.030	✓	0.0417
7	0.0400	0.035	✗	0.0571
8	0.1200	0.040	✗	0.1500



BH by hand, $m = 10$, $\alpha = 0.05$: the largest i that passes is 6, so ranks 1–6 are rejected; Bonferroni at 0.005 would reject only 2.

Double dipping: selecting on the statistic you then test with

- You filter to the 500 most variable genes, then test those 500 for differential expression
- Variability between groups IS part of what makes a gene variable overall
- So you selected genes for looking interesting, and then tested whether they look interesting
- "We only tested the top 500" is not a defence. It is the confession.



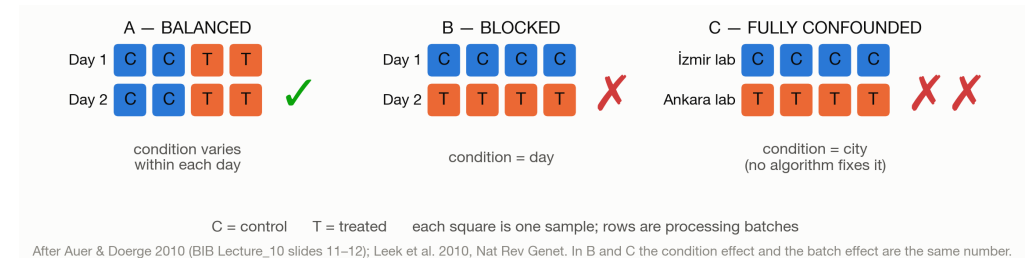
*Double dipping (top) versus an independent filter (bottom): would the filter pick the same genes if the labels were shuffled?
Kriegeskorte et al. 2009.*

Which filters are legitimate?

- Legitimate filters are independent of the group labels
- Remove genes with zero counts everywhere, or below a mean-count floor computed across ALL samples
- Illegitimate filters look at the labels, even indirectly
- It returns in Week 9 as leakage: never let the thing you are testing influence the thing you tested on

Confounding and batch: the error no algorithm fixes

- Layout A — BALANCED. Days 1 and 2 each: 2 control, 2 treated. ✓
- Layout B — BLOCKED. Day 1: 4 control. Day 2: 4 treated. ✗ (condition = day)
- Layout C — CONFOUNDED. All controls in İzmir, all treated in Ankara. ✗✗
- In B and C the condition effect and the batch effect are the same number
- Perfect confounding: no algorithm fixes it. Not ComBat. Not ~batch + condition. Not Harmony.
- The fix costs zero lira and happens before you pipette: randomise and block



Balanced, blocked, and fully confounded layouts: in B and C the condition effect and the batch effect are the same number. After Auer & Doerge 2010; Leek et al. 2010.

The GENOMES M row goes live

- Since Week 2 your GENOMES card has said M: n/a — not yet taught
- As of right now it is live, and it is required in every notebook to the end of term
- M = Multiple testing and the unit of independence. Three things, every time:
 - What is my independent unit, and what is n?
 - What correction did I apply, and to how many tests?
 - Did I filter on the same statistic I then tested with?
- Fill it on screen, now, for the notebook you are about to open

the GENOMES card as issued in Week 2 — row M goes live today

GENOMES	
seven controls · every notebook, every week	
	your entry
G	Genome build + annotation release: all inputs agree?
E	Environment: versions and the seed
N	Rows and columns, before and after every filter
O	Off-by-one: which coordinate convention?
M	Multiple testing and the unit of independence n/a until W6 unit = simulated sample (n=3 per group); BH across all 12,000 genes tested; no pre-filter applied before testing
E	Entities as stable IDs (ENSG..., not symbols)
S	Sources resolved to a DOI — and opened

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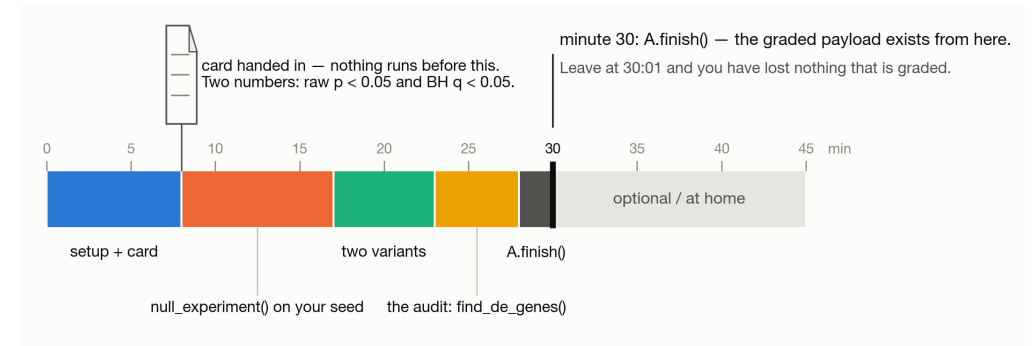
The GENOMES card from Week 2 with row M going live: unit = simulated sample (n=3 per group); BH across all 12,000 genes tested; no pre-filter applied before testing.

This afternoon: the negative control

- 45 minutes. Zero downloads except one curl of the helper module from EgeDers
- Everything is simulated from your seed — the wifi-proof lab of the semester
- Minutes 0–8: setup, and the predict-then-check card — on paper, handed in, and it is the attendance record
- Two numbers only: how many of 12,000 at raw $p < 0.05$, and how many at BH $q < 0.05$
- Nothing runs until the card is handed in. That is the one rule of the hour.

Run of show: minutes 8 to 30

- Minutes 8–17: `null_experiment()` on your seed. Row counts printed at every stage. Record `n_raw_sig`, `n_bh_sig`.
- Minutes 17–23: two one-click variants — pseudoreplication, then confounding. Record both.
- Minutes 23–28: the audit. A supplied AI-written `find_de_genes()` that runs cleanly and is wrong.
- Minute 30: `A.finish()`. Everything after that is optional and can be done at home.



This afternoon: card handed in at minute 8, `A.finish()` at minute 30, everything after is optional.

Before you go

- Tick "statistical inference" on your Skills Ledger
- Photograph the unit-of-independence rule and the four forbidden sentences
- Banked today, cashed later: unit of independence → Weeks 7, 9, 10 · double dipping → Week 9 as leakage · BH → every remaining notebook
- Two midterm item types were fixed today: the data-to-decision case, and the p-value-histogram figure item
- Everything on EgeDers. GitHub is the mirror.
- Next week: expression — counts, normalisation, and the TPM that isn't

APPENDIX — the rank-sum test

- Not taught in the 75 minutes; posted to EgeDers — read it before Week 10
- Wilcoxon rank-sum test (Wilcoxon sıra toplamı testi), also called Mann–Whitney U
- A non-parametric two-group test
- You do not compare the values. You rank all the values together, then sum the ranks in one group.
- The null: the labels are exchangeable, so every assignment of ranks to groups is equally likely

Why 3 vs 3 can never reach $p < 0.05$

- Enumeration: 3 vs 3 gives $C(6,3) = 20$ rank splits
- Only 20. You can write them all out.
 - The most extreme split — A holds ranks 1, 2, 3 — occurs once
 - So the smallest one-sided p-value at 3 vs 3 is $1/20 = 0.05$
- With 3 vs 3, a Wilcoxon test can NEVER produce $p < 0.05$ two-sided.
- Not for any data. The test is arithmetically incapable, not underpowered
- Why it is here: Week 10's lab is "the Wilcoxon trap"

All $C(6,3) = 20$ ways to split ranks $\{1, \dots, 6\}$ into two groups of three

#	ranks in A	ranks in B	sum of A	#	ranks in A	ranks in B	sum of A
1	1 2 3	4 5 6	6	11	2 3 4	1 5 6	9
2	1 2 4	3 5 6	7	12	2 3 5	1 4 6	10
3	1 2 5	3 4 6	8	13	2 3 6	1 4 5	11
4	1 2 6	3 4 5	9	14	2 4 5	1 3 6	11
5	1 3 4	2 5 6	8	15	2 4 6	1 3 5	12
6	1 3 5	2 4 6	9	16	2 5 6	1 3 4	13
7	1 3 6	2 4 5	10	17	3 4 5	1 2 6	12
8	1 4 5	2 3 6	10	18	3 4 6	1 2 5	13
9	1 4 6	2 3 5	11	19	3 5 6	1 2 4	14
10	1 5 6	2 3 4	12	20	4 5 6	1 2 3	15

The extreme splits (sum 6 or sum 15) each occur in exactly 1 of 20 arrangements: one-sided $p = 1/20 = 0.05$, two-sided $p = 2/20 = 0.10$. With 3 vs 3, a Wilcoxon rank-sum test can never give $p < 0.05$ two-sided — for any data.

All $C(6,3) = 20$ splits of six ranks into two groups of three; the extreme sums 6 and 15 each occur once, so with 3 vs 3 the rank-sum test can never reach $p < 0.05$ two-sided.